

Module 4 Java Swing Hamburger Order System Lab Assignment



Java Swing Hamburger Order System Lab Assignment

Objective:

The objective of this lab assignment is to implement a simple Java Swing application for managing hamburger orders using a MariaDB database. Students are expected to use the provided skeleton code and complete the missing functionalities to create a functional Hamburger Order System.

Task Description:

1. Database Setup (15 points):

- Implement the `createTableIfNotExists` method to create a table named `orders` with the specified columns.
- Ensure that the `order_id` is set as the primary key with auto-increment.

2. Place Order (25 points):

- Implement the `placeOrder` method to capture customer name and burger type through input dialogs.
- Insert the order details (customer name, burger type, and order date) into the `orders` table.

3. View Order History (25 points):

- Implement the `viewOrderHistory` method to retrieve and display the order history from the `orders` table.
- Display the order details in a formatted manner.

4. Cancel Order (25 points):

- Implement the `cancelOrder` method to prompt the user for an order ID to cancel.
- Delete the specified order from the `orders` table.

5. Code Structure and Readability (10 points):

- Ensure the code follows good practices, including proper indentation and meaningful variable/method names.
- Include comments where necessary to explain the functionality of methods and sections of code.

Grading Rubric:

- **Database Setup (15 points):**

- Successful creation of the `orders` table with the correct structure: 5 points.
- Primary key and auto-increment settings: 5 points.
- Proper error handling for database operations: 5 points.

- **Place Order (25 points):**

- Successful capture of customer name and burger type: 5 points.
- Correct insertion of order details into the database: 10 points.
- Proper error handling for database operations: 5 points.
- Clear and user-friendly messages: 5 points.

- **View Order History (25 points):**

- Successful retrieval and display of order history: 10 points.
- Proper formatting of displayed order details: 10 points.
- Proper error handling for database operations: 5 points.

- **Cancel Order (25 points):**

- Successful prompt for order ID and cancellation: 10 points.
- Correct deletion of the specified order from the database: 10 points.
- Proper error handling for database operations: 5 points.

- **Code Structure and Readability (10 points):**

- Consistent indentation and formatting: 5 points.
- Meaningful variable/method names: 5 points.

Submission Guidelines:

- Submit the completed Java Swing Hamburger Order System code as a single file.
- Include comments explaining the purpose of each method and major sections of code.
- Provide a brief summary of any challenges faced during implementation.

Note: Adjust the point distribution based on the specific emphasis and requirements of your course.

**** Skeleton Code ****

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.*;

public class SwingHamburgerOrderSystem {

    private static final String DB_URL = "jdbc:mariadb://localhost:3306/hamburger_db";
    private static final String DB_USER = "your_username";
    private static final String DB_PASSWORD = "your_password";

    public static void main(String[] args) {
        createTableIfNotExists();

        SwingUtilities.invokeLater(() -> {
            JFrame frame = new JFrame("Hamburger Order Management System");
            frame.setSize(400, 300);
            frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

            JPanel panel = new JPanel();
            frame.add(panel);
            placeComponents(panel);
        });
    }
}
```

```
        frame.setVisible(true);
    });
}

private static void placeComponents(JPanel panel) {
    panel.setLayout(new FlowLayout());

    JButton placeOrderButton = new JButton("Place Order");
    panel.add(placeOrderButton);

    JButton viewOrderHistoryButton = new JButton("View Order History");
    panel.add(viewOrderHistoryButton);

    JButton cancelOrderButton = new JButton("Cancel Order");
    panel.add(cancelOrderButton);

    JButton exitButton = new JButton("Exit");
    panel.add(exitButton);

    placeOrderButton.addActionListener(new ActionListener() {
        @Override
        public void actionPerformed(ActionEvent e) {
            placeOrder();
        }
    });

    viewOrderHistoryButton.addActionListener(new ActionListener() {
        @Override
        public void actionPerformed(ActionEvent e) {
            viewOrderHistory();
        }
    });
}
```

```
cancelOrderButton.addActionListener(new ActionListener() {  
    @Override  
    public void actionPerformed(ActionEvent e) {  
        cancelOrder();  
    }  
});  
  
exitButton.addActionListener(new ActionListener() {  
    @Override  
    public void actionPerformed(ActionEvent e) {  
        System.exit(0);  
    }  
});  
}  
  
private static void createTableIfNotExists() {  
    // TODO: Implement table creation logic  
}  
  
private static void placeOrder() {  
    // TODO: Implement order placement logic  
}  
  
private static void viewOrderHistory() {  
    // TODO: Implement order history viewing logic  
}  
  
private static void cancelOrder() {  
    // TODO: Implement order cancellation logic  
}  
}
```

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 12	Everyone	-	Oct 22 at 11:59pm